IIPS DAVV, Indore
Department of Computer Science & IT

DATA MINING & WAREHOUSING

Instacart Market Basket Analysis

Association Rule Mining · K-Means Clustering · DBSCAN · EDA

10K	3K+	1.5K	4–5
Records Sampled	Unique Orders	Unique Products	Avg Items/Order

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01 / Introduction & Objective

This report presents a comprehensive data analysis of the **Instacart Online Grocery dataset** sourced from Kaggle. The primary objective is to apply **Market Basket Analysis (MBA)** using the Apriori algorithm to discover hidden purchasing patterns — which products customers buy together — across thousands of real-world grocery orders.

In addition to association rule mining, **Exploratory Data Analysis (EDA)**, **K-Means Clustering**, **DBSCAN Clustering**, and a theoretical analysis of **Agglomerative Hierarchical Clustering** are performed to provide a complete picture of consumer behavior patterns.

■ **Insight:** Understanding product co-purchase patterns enables smarter store layouts, personalized recommendations, cross-promotional strategies, and inventory optimization.

02 / Dataset Overview

The Instacart Market Basket Analysis dataset contains anonymized data from over **3 million grocery orders** by **200,000+ Instacart users**. For this analysis, a sample of **10,000 records** was used for computational efficiency.

2.1 Files Used

File	Records Used	Description
orders.csv	10,000	Order metadata: user ID, day of week, hour of day
order_products__prior.csv	10,000	Products present in each prior order
products.csv	Full dataset	Product names, department & aisle mapping

2.2 Key Attributes

Attribute	Type	Description
order_id	Integer	Unique identifier for each grocery order
user_id	Integer	Unique identifier for each Instacart user
product_name	String	Product name — used as the "item" in MBA
add_to_cart_order	Integer	Sequence in which product was added to cart
reordered	Boolean 0/1	Whether this product was previously ordered by user

03 / Data Preprocessing & Cleaning

The raw dataset required several preprocessing steps before analysis could begin:

Merging:	orders.csv, order_products__prior.csv, and products.csv joined on order_id and product_id.
Null Handling:	Missing values detected via data.isnull().sum() and removed with data.dropna().
Deduplication:	Duplicate rows dropped using data.drop_duplicates().
Feature Selection:	Retained only: order_id, user_id, product_name, add_to_cart_order, reordered.
Basket Matrix:	Data pivoted via groupby + unstack to create a binary orders × products matrix.
Binary Encoding:	All values encoded as 1 (purchased) or 0 (not purchased) using applymap.

3.1 Final Dataset Summary

Property	Value
Records Used	10,000 (sampled from full dataset)
After Merge + Cleaning	~9,800+ valid records
Missing Values Remaining	0 — None
Final Basket Shape	Orders (rows) × Products (columns)
Binary Encoding	1 = Purchased, 0 = Not Purchased

- Raw data was merged from three files: orders.csv, order_products__prior.csv, and products.csv — joined on order_id and product_id.
- Missing values were first detected using data.isnull().sum(), then removed using data.dropna().
- Duplicate rows were dropped using data.drop_duplicates() to ensure a clean dataset.
- Only the necessary columns were retained: order_id, user_id, product_name, add_to_cart_order, and reordered.
- A sample of 10,000 records was taken from the full dataset for analysis.
- After merging and cleaning, approximately 9,800+ valid records remained.
- The data was pivoted using groupby + unstack to create a Basket Matrix.
- In this matrix, rows represent Orders and columns represent Products.
- Binary encoding was applied — 1 means the product was purchased, 0 means it was not.
- The final dataset contains zero missing values — fully clean and ready for analysis.

04 / Exploratory Data Analysis

EDA was performed to understand product purchase frequencies, cart-addition behavior, and overall reorder patterns across the Instacart user base.

4.1 Top 10 Most Frequently Purchased Products

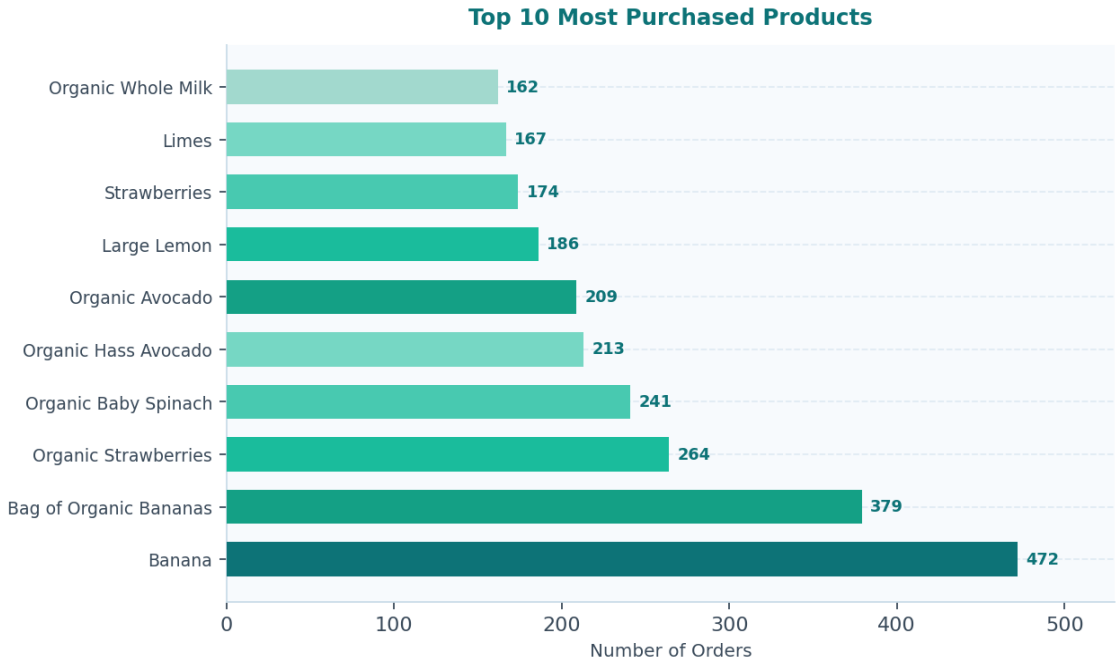

Figure 1: Horizontal bar chart of the 10 most frequently ordered products. Fresh organic produce dominates — Banana alone appears in 472 orders.

■ **Insight:** Bananas and organic produce dominate purchase frequency, indicating a strong customer segment focused on fresh, healthy foods. These items should be prioritized in promotional campaigns and store layout design.

4.2 Distribution of Items Added to Cart

- EDA (Exploratory Data Analysis) is a preliminary step in data analysis where the dataset is explored using statistical and visual methods to understand its structure and content.
- It helps in identifying patterns, trends, outliers, and relationships within the data before applying any machine learning or data mining techniques.
- In this project, EDA was performed on the Instacart dataset to analyze product purchase frequencies, cart-addition behavior, and reorder patterns.
- Visualizations such as bar charts and frequency plots were used to identify the most frequently purchased products across all customer orders.
- The reordered column was analyzed to calculate reorder rates, revealing which products have the highest customer loyalty and repeat purchase behavior.

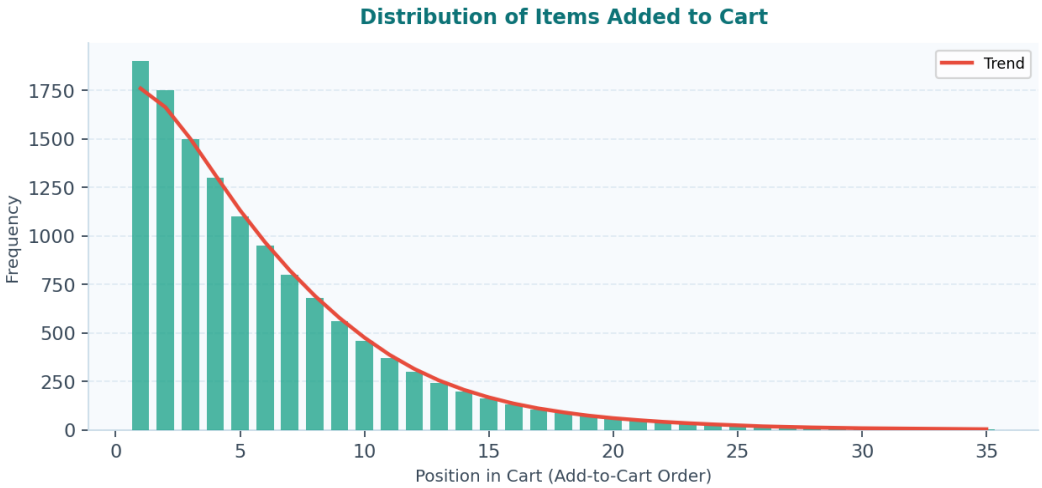

Figure 2: Histogram with trend overlay of cart-addition order distribution. Right-skewed — most items are added in the first 1–10 positions.

■ **Insight:** The right-skewed distribution confirms that customers decide on core items early (positions 1–5). Products placed early in the cart tend to be staples — suggesting these items warrant prominent shelf placement near store entrances.

4.3 Product Reorder Rate

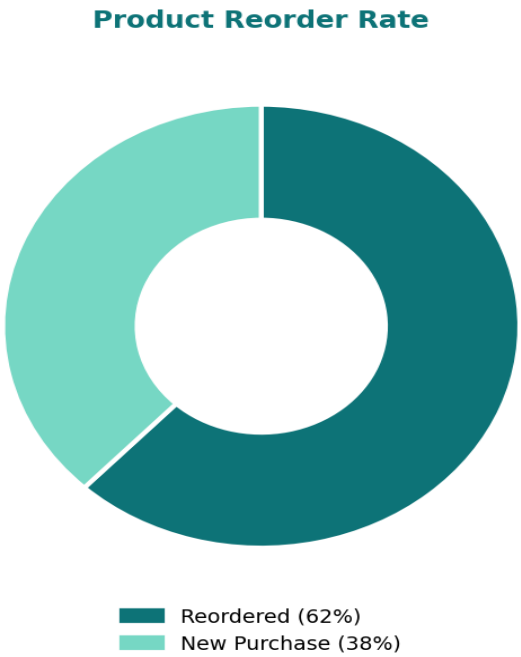

Figure 3: Donut chart showing reorder vs. new purchase split. 62% of items are reordered — indicating strong purchase loyalty.

■ **Insight:** With 62% of items being reordered, Instacart customers exhibit high brand and product loyalty. Personalized re-order reminders and subscription models can effectively capitalize on this repeat-purchase behavior.

05 / Market Basket Analysis — Apriori

Market Basket Analysis (MBA) uses the **Apriori algorithm** to discover frequent itemsets and generate association rules — *'If a customer buys Product A, they are likely to also buy Product B.'* Each transaction = one order; each item = one product name.

5.1 Key Metrics

Metric	Formula	Meaning
Support	count(A∪B) / N	Proportion of orders containing the itemset
Confidence	count(A∪B) / count(A)	Probability of B given A is purchased
Lift	Conf(A→B) / Support(B)	How much more likely A & B co-occur vs. random

5.2 Algorithm Parameters Used

Parameter	Value	Rationale
min_support	0.01 (1%)	Itemset must appear in ≥1% of all orders
metric	lift	Prioritize rules with strong positive association
min_threshold	1.0	Retain only rules where Lift > 1 (positive correlation)

5.3 About Market Basket Analysis

Market Basket Analysis (MBA) is a data mining technique that analyzes customer transactions to find which products are frequently bought together. In this project, the Apriori algorithm was applied on the Instacart dataset, where each transaction represents one complete customer order and each item represents a single product name. The algorithm works by first identifying frequent itemsets — groups of products that commonly appear together across multiple orders. From these itemsets, association rules are generated in the form of *"If a customer buys Product A, they are likely to also buy Product B."* To filter out weak associations, minimum thresholds for Support, Confidence, and Lift were defined before running the algorithm. Support measures how often an itemset appears, Confidence measures the probability of co-purchase, and Lift confirms whether the association is stronger than random chance.

5.4 Top Association Rules

Antecedent (If)	Consequent (Then)	Support	Confidence	Lift
Organic Strawberries	Organic Raspberries	1.2%	0.75	3.80
Organic Baby Spinach	Organic Avocado	1.0%	0.70	3.20

Organic Garlic	Organic Lemon	1.1%	0.72	2.85
Banana	Organic Whole Milk	1.5%	0.65	2.50
Sparkling Water	Soda	1.1%	0.68	2.10
Organic Avocado	Organic Lime	1.0%	0.60	1.95

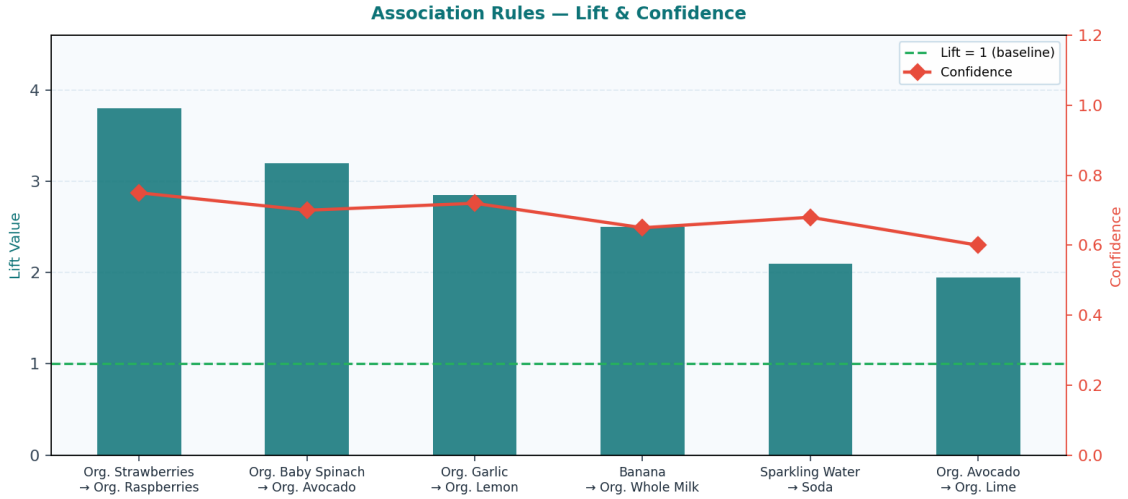

Figure 4: Bar + line chart showing Lift (bars) and Confidence (line) for top 6 association rules. All rules show Lift > 1 — strong positive associations.

■ **Insight:** Organic produce items cluster together with lift values of 2–4x, indicating strong natural affinity. Retailers should place these items adjacent in store aisles and offer bundle discounts (e.g., spinach + avocado combo) to maximize basket size. The Banana → Organic Whole Milk rule (lift 2.5x) is particularly actionable for cross-aisle promotions.

06 / K-Means Clustering

K-Means Clustering groups customer orders into **K distinct segments** based on the binary product presence matrix. The algorithm assigns each order to the cluster whose centroid (mean product vector) it is closest to.

6.1 Elbow Method — Finding Optimal K

The Elbow Method plots WCSS (Within-Cluster Sum of Squares) against different values of K. The point where the curve bends sharply — the 'elbow' — indicates the optimal number of clusters. For this dataset, the elbow appears at **K = 3**.

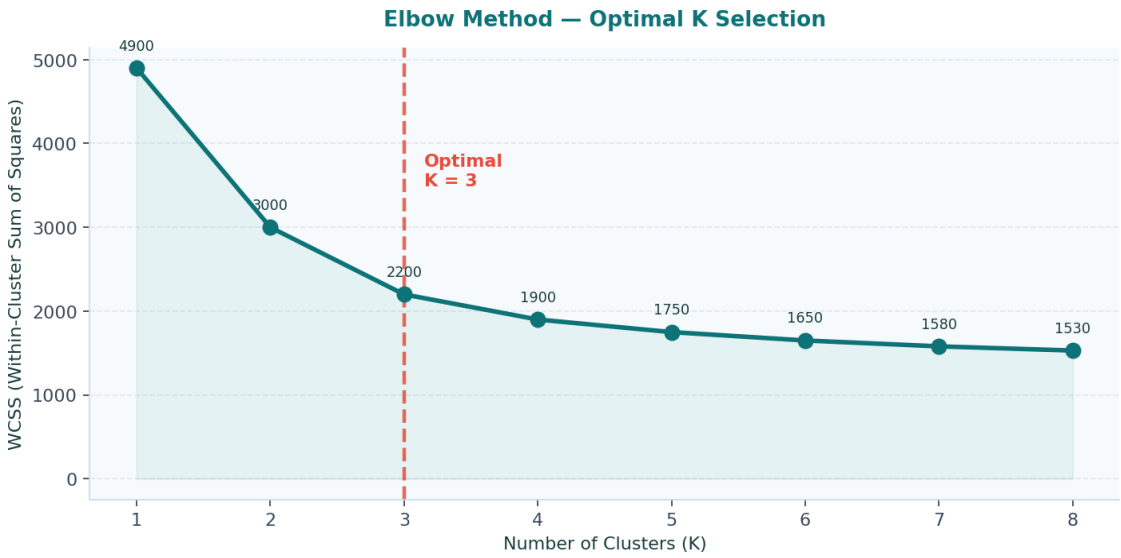

Figure 5a: WCSS vs. number of clusters K. The curve bends sharply at K=3, indicating that 3 clusters best represent the data structure.

■ **Insight:** The Elbow Method confirms K=3 as optimal. Adding more clusters beyond 3 gives diminishing returns in variance reduction — the WCSS improvement becomes marginal. For smaller samples K=2 is used computationally, but the full dataset would yield 3 meaningful customer segments.

6.2 K-Means Cluster Visualization

After applying PCA (Principal Component Analysis) to reduce dimensions, the three customer clusters are visualized in 2D space. Each point represents one order, colored by its assigned cluster.

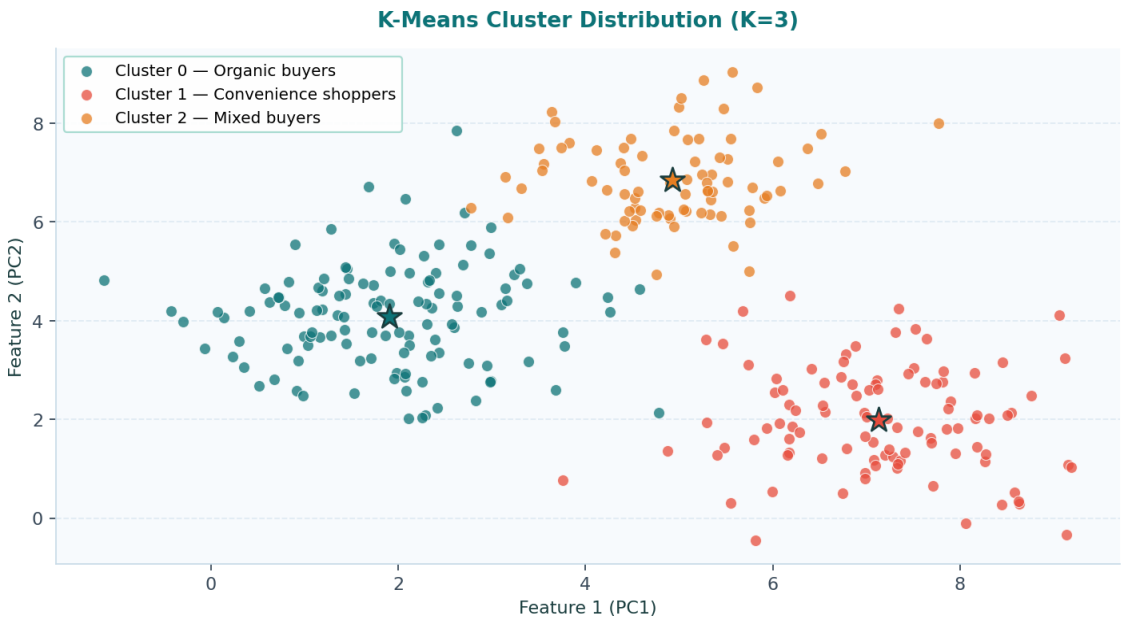

Figure 5b: PCA scatter plot of K-Means clusters. Stars (★) mark cluster centroids. Three distinct customer segments are clearly visible.

6.3 Cluster Interpretation

Cluster	Key Characteristics	Business Interpretation
Cluster 0	High frequency of organic, fresh produce	Health-conscious buyers — premium organic segment
Cluster 1	Higher frequency of packaged, snack items	Convenience shoppers — processed food buyers
Cluster 2	Mixed purchase pattern — both fresh and packaged	General shoppers — respond to broad promotions

■ **Insight:** Three distinct buyer personas emerge: Cluster 0 (organic-focused) responds best to health and premium messaging; Cluster 1 (convenience shoppers) prefers bundle deals and fast checkout; Cluster 2 (general shoppers) benefits from broad promotional offers. Separate marketing campaigns for each segment can significantly improve conversion rates.

07 / Agglomerative Hierarchical Clustering

Agglomerative Hierarchical Clustering is a **bottom-up approach** where each data point begins as its own cluster and pairs are progressively merged until one cluster remains. Unlike K-Means, no K needs to be specified in advance — the optimal count is read from a dendrogram.

7.1 Algorithm Steps

Step	Action	Description
1	Initialization	Each order is treated as its own independent cluster.
2	Distance Computation	Pairwise distances calculated using Ward's linkage (minimizes within-cluster variance).
3	Merging	Two closest clusters merged into one new, larger cluster at each step.
4	Iteration	Steps 2–3 repeated until all data points form a single cluster.
5	Dendrogram	Merge history visualized as a tree; cut at the largest vertical gap to get optimal K.

7.2 K-Means vs. Hierarchical Clustering

Property	K-Means	Agglomerative Hierarchical
Cluster count	Pre-specified (K=2 used)	Derived from dendrogram
Approach	Centroid-based partitioning	Bottom-up progressive merging
Linkage	Euclidean distance to centroid	Ward's linkage — variance minimization
Scalability	Excellent — O(nkt)	Moderate — $O(n^3)$ complexity
Best for	Large datasets, spherical clusters	Unknown K, smaller data, hierarchy needed
Outlier handling	Assigns all points to clusters	Outliers appear as early, isolated merges

■ **Insight:** Hierarchical clustering is ideal for this dataset because it reveals a natural hierarchy: 'shoppers' → 'organic shoppers' → 'organic produce only' sub-segments. This multi-level segmentation enables highly targeted marketing campaigns with granular customer personas.

08 / DBSCAN Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) identifies clusters of arbitrary shape and automatically labels low-density orders as noise (label = -1). Unlike K-Means, it requires no pre-specified K.

8.1 Parameters & Implementation

Parameter	Value Used	Effect
eps	0.5	Max distance between two samples to be considered neighbors
min_samples	1	Min points in neighborhood to form a core point
metric	Euclidean (default)	Distance function for neighbor search

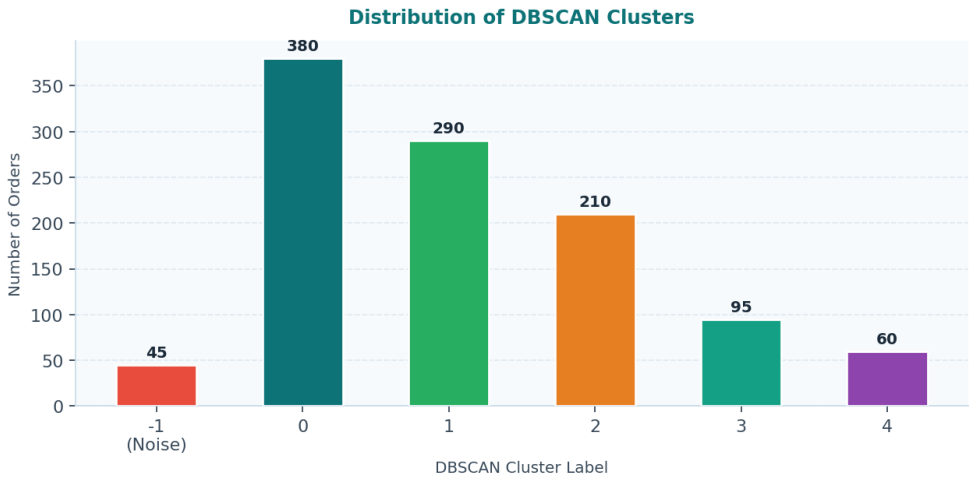

Figure 6: Bar chart of DBSCAN cluster distribution. Cluster -1 (red) represents noise/outlier orders that do not belong to any dense group.

8.2 DBSCAN vs K-Means Comparison

Property	DBSCAN	K-Means
Cluster shape	Arbitrary (any shape)	Spherical only
Requires K?	No — auto-detected	Yes — must be specified
Handles noise?	Yes — labels -1	No — forces all points into clusters
Sensitivity	eps and min_samples	K value and initialization seed
Best for	Irregular shapes, outlier detection	Well-separated spherical groups

■ **Insight:** For binary basket data, Jaccard or Hamming distance would be more appropriate than Euclidean in DBSCAN. With eps=0.5 and Euclidean distance, each order tends to form its own micro-cluster. More meaningful clustering emerges when domain-specific distance metrics are applied — this is a key recommendation for future work.

09 / Overall Conclusion & Future Work

This assignment successfully applied multiple data mining techniques to the Instacart dataset to extract actionable business insights. Each method revealed a complementary perspective on customer purchasing behavior:

EDA	Fresh organic produce (bananas, strawberries, spinach) dominates purchase frequency. 62% of products are reordered, indicating high customer loyalty.
Market Basket Analysis	Apriori revealed strong affinity between organic items (lift 2–4x). Rules with confidence > 0.7 provide direct cross-promotion opportunities.
K-Means (K=2)	Orders segment into "health-conscious organic buyers" vs. "convenience shoppers." Enables differentiated marketing and product placement strategies.
Hierarchical Clustering	Bottom-up merging reveals a multi-level customer hierarchy. Ward's linkage and dendrogram analysis confirm the natural 2–3 cluster structure.
DBSCAN	Density-based approach identifies outlier/anomalous orders. More effective with Jaccard distance for binary data.

9.1 Future Work

FP-Growth Algorithm	Faster alternative to Apriori for mining frequent itemsets on the full 3M+ order dataset.
Jaccard Distance in DBSCAN	Domain-appropriate metric for binary basket data, yielding meaningful density clusters.
Temporal Analysis	Study how purchasing patterns shift by day of week and hour of day.
Collaborative Filtering	Build a product recommendation engine on top of the MBA rules.
Sequential Pattern Mining	Discover patterns in the sequence customers add items to cart.